Walk calibration of BTL modules in testbeam:

Via Synchronization of MCP scope data with FEBd

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Discussion draft; not approved for public release.

Purpose: provide a reliable time reference for BTL analyses

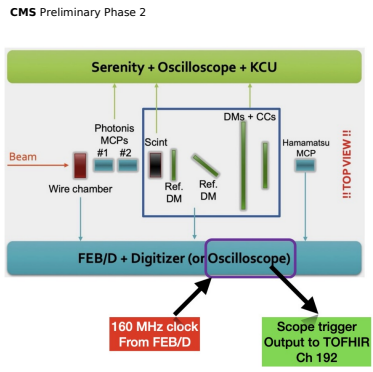
Several BTL test-beam analyses need a reliable particle-arrival time reference.

For a subset of the data, a digital oscilloscope recorded MCP-PMT signals in parallel with the BTL readout.

Approach: synchronize the two streams, use the MCP to calibrate upstream reference bars, then use their corrected times as the reference for downstream timing.

Final check: the time difference between downstream and reference bars before and after walk correction.

Parallel MCP and BTL streams are linked through FEBd

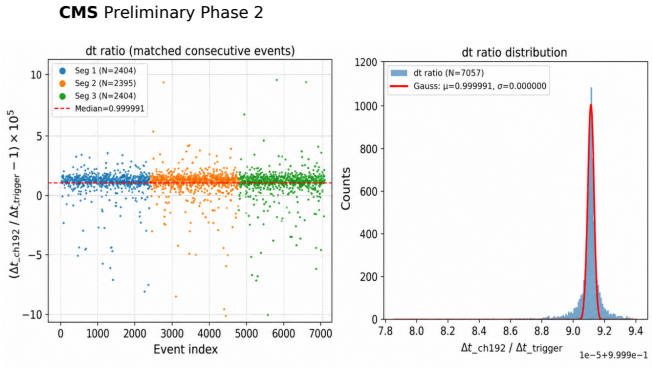


Two readout streams
The scope records the MCP-PMT pulse. FEBd records the BTL SiPM signals.

Synchronization link
The scope trigger also reaches TOFHIR channel 192 in FEBd. This link connects independently clocked records of the same particle.

September 2025 beam test. The highlighted lower path provides the reference-bar data. This FEBd-based synchronization is specific to the test-beam setup, not the final HL-LHC synchronization scheme.

Events are correlated using the intervals between triggers

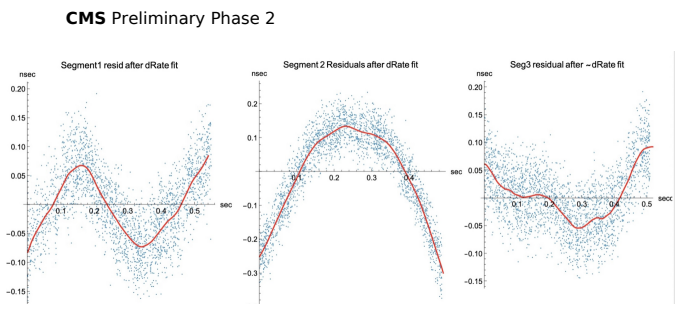


1. Compare consecutive trigger intervals in channel 192 and the scope.
2. Find consistent interval patterns.
3. Keep only unique event matches.

Consistent trigger patterns identify coincident events in the two streams.
Event matching alone does not remove relative clock drift.

The correlated intervals and their ratio test event matching, not detector resolution. GitHub: trigger matching.

The matched clocks show low-frequency relative drift



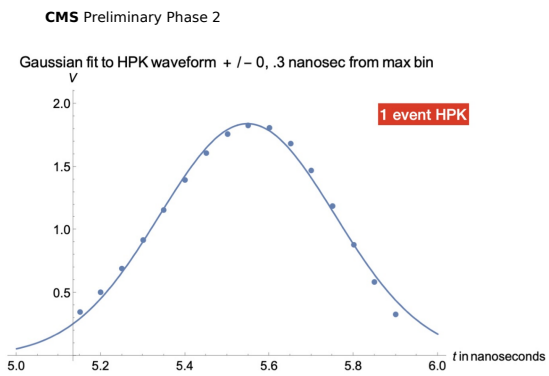
Correct the timing coordinate

1. Split data at clock resets.
2. Fit a line between matched trigger times in each segment.
3. Fit the remaining drift with LOWESS.

Add the drift to the line prediction to obtain $g(u)$, the expected TOFHIR trigger time.

Points show residuals before LOWESS; red curves describe smooth drift. Reject groups with pre-LOWESS residual scale $1.4826\,\mathrm{MAD} > 1$ ns. GitHub: clock reconstruction.

After synchronization, the MCP defines the calibration time



Fit the MCP waveform to estimate its pulse time.
Within the scope waveform, define

$$\delta_{\mathrm{MCP}} = t_{\mathrm{peak}} - t_{\mathrm{trigger}}.$$

For SiPM channel i, use

$$\Delta t_i = t_i - g(u) - \delta_{\mathrm{MCP}}.$$

The charge-dependent part of this difference is the walk.

The waveform and fitted template determine the MCP pulse time. u is the scope-trigger timestamp. A fixed cable/geometry offset is irrelevant to the shape; clock-rate conversion over the short waveform interval is neglected.

Measure the upstream channels' time walk versus charge

Separate threshold settings before selection and fitting.

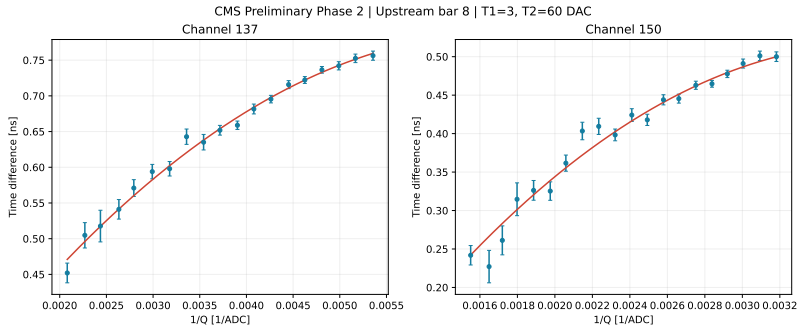
1. In run 4399, require exactly five TOFHIR hits and a unique MCP match: upstream channels 137 and 150, downstream channels 234 and 243, and channel 192.
2. Within each setting, require clock quality and finite positive charge, ΔI and threshold-rise time. Keep charge above the spectrum's left half-maximum boundary.
3. Form 20 uniform $1/Q$ bins. Clip timing outliers at $5(1.4826\,\mathrm{MAD})$ from each bin's median; require at least five entries.
4. Fit the bin means with weights $1/\mathrm{SEM}^2$:

$$W_i(Q) = a_i + b_i/Q + c_i/Q^2.$$

Q is the measured ADC charge proxy, not energy in MeV. SEM denotes the standard error of the mean.

Upstream bar 8 and downstream bar 3. Combine files only at identical scan settings. GitHub: selected reference sample.

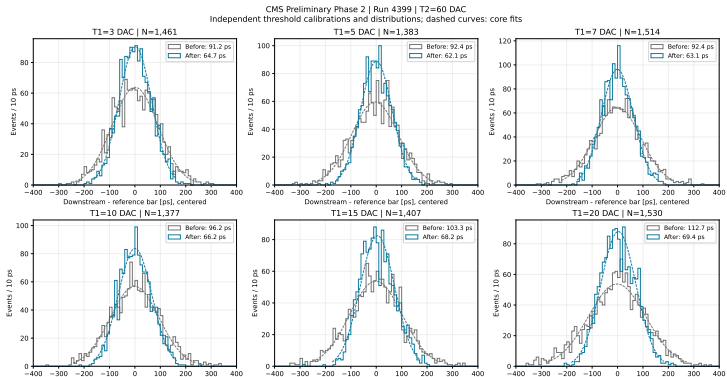
Walk-corrected upstream bars provide the time reference



Correct each end: $t_i^{\mathrm{corr}} = t_i - W_i(Q_i)$. Use the bar average: $t_{\mathrm{ref}} = (t_L^{\mathrm{corr}} + t_R^{\mathrm{corr}})/2$.

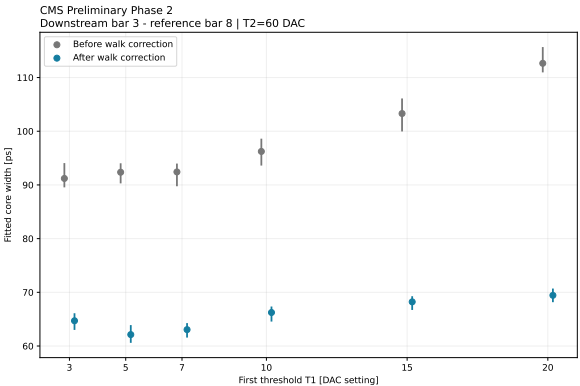
Upstream bar 8, lowest scanned setting: $T_1 = 3$, $T_2 = 60$ (DAC settings). Points show profile means with SEM; curves are SEM-weighted quadratics in $1/Q$. The $+26$ ns display offset does not change the shape. Separate fits are made for every setting and all four channels.

Measure bar-to-bar jitter separately at each threshold



Downstream bar 3 minus upstream reference bar 8. Within each panel, the same events are used before and after four-channel correction, inside populated charge support. Histograms are median-centered for display only; no per-file offsets are removed. Dashed curves are truncated-normal core fits within median $\pm 2.5(1.4826\,\mathrm{MAD})$.

Final check: timing improves at fixed operating points



One point per threshold
Each setting has its own charge selection and four walk corrections.
Compare the two corrected bar-average times. The shared MCP time cancels.
Do not pool thresholds. Only the separately fitted widths enter an optional combined summary.

Vertical intervals: 16th–84th percentiles from 400 paired file/segment-block bootstraps per setting, with fixed selection and calibrations. This is in-sample validation; calibration uncertainties and fit-window dependence are not included. The pair width does not by itself determine either bar's resolution.

References

[1] GitHub repository, `https://github.com/De-Cristo/LecroyTranslator4CMSTimingDetector`.

[2] Presentation by the Saclay group. `https://indico.cern.ch/event/1619706/`

[3] "Timing Performance of a Micro-Channel-Plate Photomultiplier Tube", `https://arxiv.org/pdf/1909.12604`

[4] CMS Collaboration, "A MIP Timing Detector for the CMS Phase-2 Upgrade". Technical Report CERN-LHCC-2019-003. CMS-TDR-020, CERN, Geneva, 2019 and "A High-Granularity Timing Detector for the ATLAS Phase-II Upgrade", CERN-LHCC-2020-007; ATLAS-TDR-031

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[6] W. R. Leo, "Techniques For Nuclear and Particle Physics Experiments", p.326, Springer-Verlag Berlin Heidelberg.

[7] S. White and A. Boletti, "Amplitude Walk in Fast Timing: The Role of Dual Thresholds" 2026 JINST V21 P02001.

[8] W. S. Cleveland, "Robust Locally Weighted Regression and Regression Scatterplots", J. Amer. Statist. Assoc. **74** (1979) 829.

Method for Walk Correction using Leading Edge Slope:

Derived from dual threshold feature of TOFHIR2C

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Abstract

Two discriminator thresholds measure a finite-interval slope on the pulse's leading edge. A linear fit describes the mean slope as a function of charge. Its inverse provides a time-walk model with one amplitude, AWC, per channel.

This study measures slopes in run 4076 and determines AWC by comparison with independent MCP calibration curves from run 4399. The fit uses only charge intervals populated in both runs.

The comparison tests calibration shapes, not corrected-event timing resolution. The present AWC calibration still needs the MCP reference; reference-free determination remains a future step.

Two thresholds measure how fast the pulse rises

Problem: a fixed threshold gives an amplitude-dependent timestamp. A steeper leading edge generally crosses the threshold earlier.

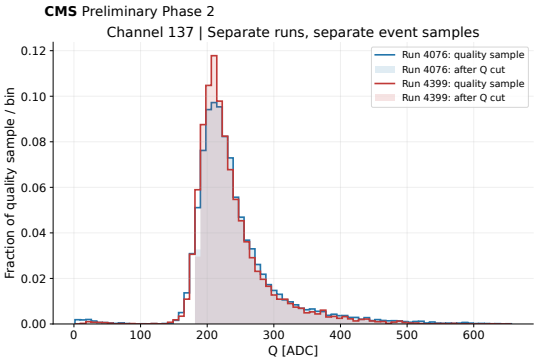
$$I_1 < I_2, \qquad t_1 < t_2, \qquad S = \frac{I_2 - I_1}{t_2 - t_1} \quad [\mu\mathrm{A/ns}].$$

1. Reconstruct both threshold-crossing times with consistent coarse and fine timing.
2. Divide the threshold-current separation by the crossing-time separation.
3. Use charge Q to describe the mean slope for each channel.

S is a finite-interval proxy, not the instantaneous derivative or the current difference itself.

Run 4076 uses $\Delta I = 7.8125\,\mu$A from the implemented threshold conversion. GitHub: timing and threshold reconstruction.

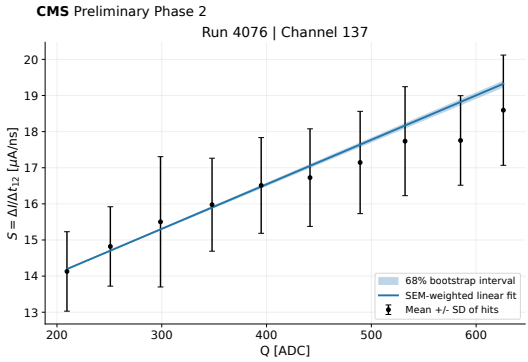
1. Select the prompt population and the charge peak region



1. Require one hit per target-bar end; no other hit in that module.
2. Leave the other module unconstrained; ignore channel 192.
3. Require finite Q, S, positive threshold and time separations, and the same coarse bin.
4. Keep $Q \geq Q_{\mathrm{left}}$.

Channel 137: lines show quality-selected charge spectra; shading adds the charge selection. Q_{left} is the smoothed histogram's left half-height boundary, not a Landau-fit boundary. Run 4076 starts at about 184.9 ADC; no extra upper-charge or 100 ps minimum-rise cut is applied.

2. Fit the mean slope as a function of charge



Group the selected run-4076 hits into ten charge bins.
Fit profile means with SEM weights:

$$\overline{S}(Q) = c_0 + c_1 Q.$$

Use all selected hits, without a training/test split.
Resample hits and refit 1000 times with fixed bin edges.

Channel 137 has 19 126 selected hits. Points show mean slope; bars show the hit standard deviation (SD). SEM is the standard error of the mean. The band is the conditional 68% bootstrap interval of the fitted mean. Linearity remains a working approximation.

3. Convert the fitted slope into a walk correction

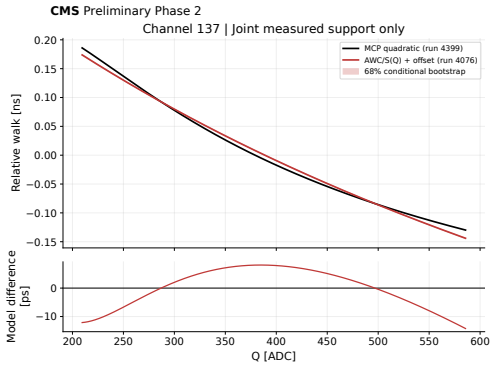
$$W_Q(Q) = \frac{A}{c_0 + c_1 Q}, \qquad t_{\mathrm{corr}} = t_{\mathrm{measured}} - W_Q(Q).$$

- A is AWC, a current amplitude in μA.
- A is constant for one channel and operating configuration.
- The correction changes event by event through Q.

The slope fit determines the charge dependence, but does not determine AWC. The present calibration obtains AWC from an external MCP reference. Use the model only over its calibrated charge range with a positive slope denominator.

The model concerns the mean pulse slope. Its band is not a single-hit timing uncertainty.

4. Determine AWC where both runs have measured charge support



Run 4076: slope model.
Run 4399: MCP quadratic $W_{\mathrm{MCP}} = a + b/Q + c/Q^2$.
Fit A and a free offset B to match

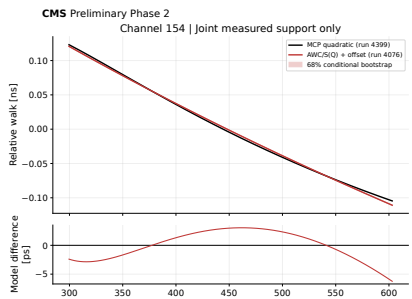
$$\frac{A}{c_0 + c_1 Q} + B$$

to the MCP curve.
Use a uniform charge grid within their populated overlap only.

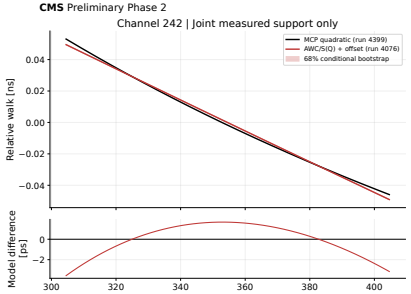
Channel 137: black = MCP reference; red = fitted slope model. The runs contain different events and assume the same nominal setup. AWC = 18.332 μA. The lower panel shows model differences, not event times; RMS mismatch = 6.80 ps.

Apply the same comparison to both detector modules

Upstream: channel 154

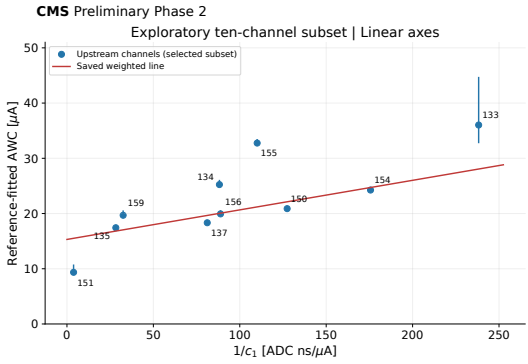


Downstream: channel 242



Each fit uses a separate AWC and offset over the measured overlap. Bands propagate run-4076 slope replicas with the MCP curve fixed. In run 4399, downstream channels require upstream bar 8 in coincidence. Run 4076 leaves the opposite module unconstrained. The opposite module does not supply a timing reference for this fit.

AWC varies between channels beyond a simple slope-coefficient trend



Each point represents one of ten selected upstream channels.
Horizontal coordinate: $1/c_1$.
Vertical coordinate: MCP-calibrated AWC.
The fitted line gives $\chi^2/\mathrm{ndf} = 538.6/8$.
A straight line does not adequately describe this illustrative subset.

Bars are slope-only 16–84% bootstrap intervals. The fit treats x as fixed and uses half the AWC interval as its symmetric error. Reference errors and correlations are omitted. The ten channels were selected for discussion, not by a derived quality rule.

The slope gives the shape; the present AWC still uses MCP

1. Select prompt hits and fit the mean slope versus charge.
2. Form the inverse-slope walk model.
3. Fit AWC and an offset to MCP curves within measured overlap.
4. Apply the charge-dependent correction within the calibrated range.

Uncertainties: displayed bands propagate slope-fit statistics with the MCP curve fixed. They exclude reference-fit, model, selection and cross-run uncertainties.

Next step: validate corrected-event timing and determine AWC from internal information. Shape agreement alone does not demonstrate reference-free calibration.

The full dual-threshold note includes the additional MCP-covariance and charge-weighting checks. GitHub: selection and profiles; GitHub: MCP reference sample.

References

[1] "Performance of the ALICE Time-Of-Flight detector at the LHC", Eur. Phys. J. Plus (2013) 128: 44

[2] "PID performance of the ALICE-TOF detector at Run 2", N. Jacazio in Proceedings of LHCP2018, https://pos.sissa.it/321/232/.

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[6] S. White and Licheng J. Zhang, "September 2025 BTL Testbeam Reference Bar Walk Calibration from Hamamatsu MCP-PMT ",cms note.

[7] E. Albuquerque et al., "TOFHIR2: The readout ASIC of the CMS Barrel MIP Timing Detector", arxiv:2404.01208

[8] J.E. Grund, SLAC-PUB-2416 (Oct. 1979).

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[10] This will arise in LGAD that are not thinned to remove the effect (J. Vavra, private communication). See also PICOSEC, "Fast Timing with MicroPattern Gas Detectors", where charge in the electron peak is captured-ignoring the ion tail..

[11] W. S. Cleveland, "Robust Locally Weighted Regression and Regression Scatterplots", J. Amer. Statist. Assoc. **74** (1979) 829.

[12] GitHub repository, https://github.com/De-Cristo/LecroyTranslator4CMSTimingDetector.